

The axioms of n -exact categories

BACHELOR'S THESIS

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Abstract

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Introduction

0.1 Prerequisites

In order to understand this paper, the reader should have some knowledge about modern homological algebra. In particular, they should already be familiar with category theory, more specifically abelian category theory, as is presented for example in the lecture notes of MA3204 Homological Algebra [[1](#), Chapter 5]

Conventions and notations

Chapter 1

An Introduction to Exact Categories

1.1 Motivation

When studying abelian categories, there are many convenient properties related to monics and epics. Let us recall a few:

Proposition 1.1 *In an abelian category, the following properties hold:*

- *the pushout of a monic is a monic;*
- *the pullback of a monic is a monic;*
- *the pushout of an epic is an epic;*
- *the pullback of an epic is an epic.*

Another nice property is the following:

Proposition 1.2 *In an abelian category,*

- every monic is a kernel of its cokernel;
- every epic is a cokernel of its kernel.

This allows us to conclude many properties about kernels and cokernels:

Proposition 1.3 *In an abelian category, the class of kernels and the class of cokernels are closed under isomorphism. Furthermore, the following hold:*

- [E0] *For any object $C \in \mathcal{C}$, the identity morphism 1_C is a kernel;*
- [E0^{op}] *For any object $C \in \mathcal{C}$, the identity morphism 1_C is a cokernel;*
- [E1] *The class of kernels is closed under composition;*
- [E1^{op}] *The class of cokernels is closed under composition;*
- [E2] *The pushout of a kernel along any morphism is a kernel;*
- [E2^{op}] *The pullback of a cokernel along any morphism is a cokernel.*

Proof It is clear that kernels and cokernels are closed under isomorphism.

E0 and E0^{op} follow directly from the fact that the identity morphism is both the kernel and cokernel of the zero map.

We can deduce E1 and E1^{op} by recalling that composing monics (respectively epics) gives us monics and that monics are kernels of their cokernel (respectively epics are cokernels of their kernel).

As for E2, recall that kernels are always monic. Thus, by 1.1, the pushout of a kernel is monic as well. But then, by 1.2, it is a kernel of its cokernel, thus it is a kernel. The proof for E2^{op} is similar. $\square$

1.2 Exact Categories

Based on the observations from the last section, one can define a more general notion that encapsulates the idea of short exact sequences,

namely exact categories. To do so, let us first define kernel-cokernel pairs.

Definition 1.5 Let $\mathcal{A}$ be an additive category. A *kernel-cokernel pair* (i, p) in $\mathcal{A}$ is a pair of composable morphisms $A \xrightarrow{i} B \xrightarrow{p} C$ such that i is a kernel of p and p is a cokernel of i .

In the context of abelian categories, we notice that kernel-cokernel pairs indeed give rise to short exact sequences in the usual sense. In a general additive category, however, we cannot say the same since we have no guarantee that kernels and cokernels even exist for arbitrary morphisms, and thus we are unsure that $\text{Im } i$ exists. But we will use our intuition from abelian categories to expand the notion of short exact sequences to additive categories. First, let us recall what it means for a class to be closed under isomorphism.

Definition 1.6 Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$. We call a morphism $A \xrightarrow{i} B$ an *admissible monic* (or *inflation*) if there exists a morphism $B \xrightarrow{p} C$ such that $(i, p) \in \mathcal{E}$. Dually, we call a morphism $B \xrightarrow{q} C$ an *admissible epic* (or *deflation*) if there exists a morphism $A \xrightarrow{j} B$ such that $(j, q) \in \mathcal{E}$. We use $A \xrightarrow{i} B$ for admissible monics, and $B \xrightarrow{p} C$ for admissible epics. We say $\mathcal{E}$ is closed under isomorphism if whenever we have $A \xrightarrow{i} B \xrightarrow{p} C \in \mathcal{E}$ and the following commutative diagram

$$\begin{array}{ccccc}
 A & \xrightarrow{i} & B & \xrightarrow{p} & C \\
 \downarrow a & & \downarrow b & & \downarrow c \\
 A' & \xrightarrow{j} & B' & \xrightarrow{q} & C'
 \end{array}$$

with a , b , and c isomorphisms, then $(j, q) \in \mathcal{E}$.

It is usually convenient to require being closed under isomorphism, as objects with universal properties are usually unique up to isomorphism. In particular, recall that all kernel objects of a specific morphism are isomorphic. In the case of a class of kernel-cokernel pairs $\mathcal{E}$, being closed under isomorphism implies that all kernels of an admissible epic are admissible monics. Indeed, consider $B \xrightarrow{p} C$ an admissible epic and $A \xrightarrow{i} B$ an admissible monic such that $(i, p) \in \mathcal{E}$. Let $A' \xrightarrow{j} B$ be another kernel of p and $A \xrightarrow{a} A'$ be the isomorphism between both kernels. We can construct the following commutative diagram

$$\begin{array}{ccccc}
 A & \xrightarrow{i} & B & \xrightarrow{p} & C \\
 \downarrow a & & \downarrow 1_B & & \downarrow 1_C \\
 A' & \xrightarrow{j} & B & \xrightarrow{p} & C
 \end{array}$$

Since all vertical arrows are isomorphisms, we can conclude that $(j, p) \in \mathcal{E}$. Dually, we can show that all cokernels of an admissible monic are admissible epics.

We can now proceed to define exact categories, basing our definition on the properties we established for abelian categories:

Definition 1.7 Let $\mathcal{A}$ be an additive category. An *exact structure* on $\mathcal{A}$ is a class $\mathcal{E}$ of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphism and satisfying the following axioms:

- [E0] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an admissible monic;
- [E0^{op}] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an admissible epic;
- [E1] The composition of two admissible monics yields an admissible monic;
- [E1^{op}] The composition of two admissible epics yields an admissible epic;

- [E2] The pushout of an admissible monic along any morphism yields an admissible monic
- [E2^{op}] The pullback of an admissible epic along any morphism yields an admissible epic

The last two axioms can be summarized by the following commutative diagrams:

$$\begin{array}{ccc}
 C & \xrightarrow{\quad f' \quad} & D \\
 \downarrow c & \text{PO} & \downarrow d \\
 C' & \xrightarrow{\quad f \quad} & D'
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 C' & \xrightarrow{\quad f' \quad} & D' \\
 \downarrow c & \text{PB} & \downarrow d \\
 C & \xrightarrow{\quad f \quad} & D
 \end{array}$$

If $\mathcal{E}$ is an exact structure on $\mathcal{A}$, we say $(\mathcal{A}, \mathcal{E})$ is an *exact category* and we call the elements of $\mathcal{E}$ *short exact sequences* (or *conflations*).

Chapter 2

Expanding to n -exact categories

Chapter 3

Generalizing results from
exact to n -exact categories

Bibliography

- [1] Theo Bühler. Exact categories. *Exposition. Math.*, 28(1):1–69, January 2010. [3](#)